

M12B · mcp showcase

PowerShell MCP + DrawIO MCP

InfraOps Demo

Introspect the system, then draw it

A Tool Is Just a Script

```
# InfraOps\Public\Get-InfraServiceHealth.ps1
# Comment-based help → tool description
# ValidateSet → input schema
# MaxResults → context-window cap

> Get-InfraServiceHealth -Status Running -NamePattern 'w*' -MaxResults 5
```

Name	DisplayName	Status	StartType
Winmgmt	Windows Management	Running	Automatic
WSearch	Windows Search	Running	Automatic

Any PowerShell you've already written is candidate material.

The Reveal: BareBoneServer.ps1

An MCP server is a **JSON-RPC loop on stdin/stdout**.

```
# ~40 lines of plain PowerShell. Zero dependencies.  
# No MCPServerPS, no SDK.  
  
function Get-MyPublicIP { ... }  
function Send-Response($id, $result) { ... }  
  
while ($null -ne ($line = [Console]::In.ReadLine())) {  
  switch ($req.method) {  
    'initialize' { Send-Response ... }  
    'tools/list' { Send-Response ... }  
    'tools/call' { Get-MyPublicIP; Send-Response ... }  
  }  
}
```

Now you understand what MCPServerPS does: wraps the loop, validates schemas, handles lifecycle.

MCPServerPS Does the Heavy Lifting

Dongbo Wang's **MCPServerPS** module wraps the JSON-RPC loop + schema validation + lifecycle.

```
# mcp-config.example.json
"command": "pwsh",
"args": ["-NoProfile", "-Command",
"MCPServerPS\\Start-MyMCP -Module '.\\InfraOps\\InfraOps.psd1'"]
```

In the MCP client: **four InfraOps tools** appear in the tool list.

Credit: MCPServerPS by Dongbo Wang (github.com/daxian-dbw/MCPServerPS)

Scoped Agent: InfraOps Inspector

Agents don't have to be all-or-nothing. This agent only sees **InfraOps tools**.

```
# .github/agents/infraops-inspector.agent.md
tools: [Get-InfraDiskSpace, Get-InfraServiceHealth,
Get-InfraRemoteServiceStatus, Get-InfraRecentEventError]
```

In Copilot chat:

```
@InfraOps Inspector
Are any drives running low and are there any recent system errors I should know about?
```

Agent calls Get-InfraDiskSpace + Get-InfraRecentEventError, answers the question.

Security Boundary: The Schema IS the Boundary

ValidateSet on `-ComputerName` is the **allow-list**.

```
> Check whether the Spooler service is running on lab-dc01.  
  
# Schema rejects lab-dc01 (not in allow-list)
```

"The LLM decides when to call. You decide what is callable."

```
> $env:INFRAOPS_ALLOWED_COMPUTERS = 'lab-dc01'  
# Restart MCP server, re-run prompt  
# Remote call goes through
```

Patterns to Take Home

- **Comment-based help** → tool description the LLM sees
- **ValidateSet and patterns** → input schema and security boundary
- **MaxResults and time windows** → context-window discipline

Any PowerShell you've already written is candidate material.

"The LLM decides when to call. You decide what is callable."

DrawIO MCP: Same Backbone, Different Output

PowerShell MCP introspects the system. **DrawIO MCP** visualizes it.

Prompt: `@drawio create a diagram of the InfraOps architecture`

WHAT DRAWIO MCP DOES

- Generates .drawio file
- Boxes for services, hosts, resources
- Arrows showing relationships

OPENS IN

- VS Code (DrawIO extension)
- Draw.io web editor
- Any DrawIO-compatible tool

Same MCP protocol. Completely different output domain.

MCP isn't just for code — it's a vocabulary.

The Combo: Introspect → Draw

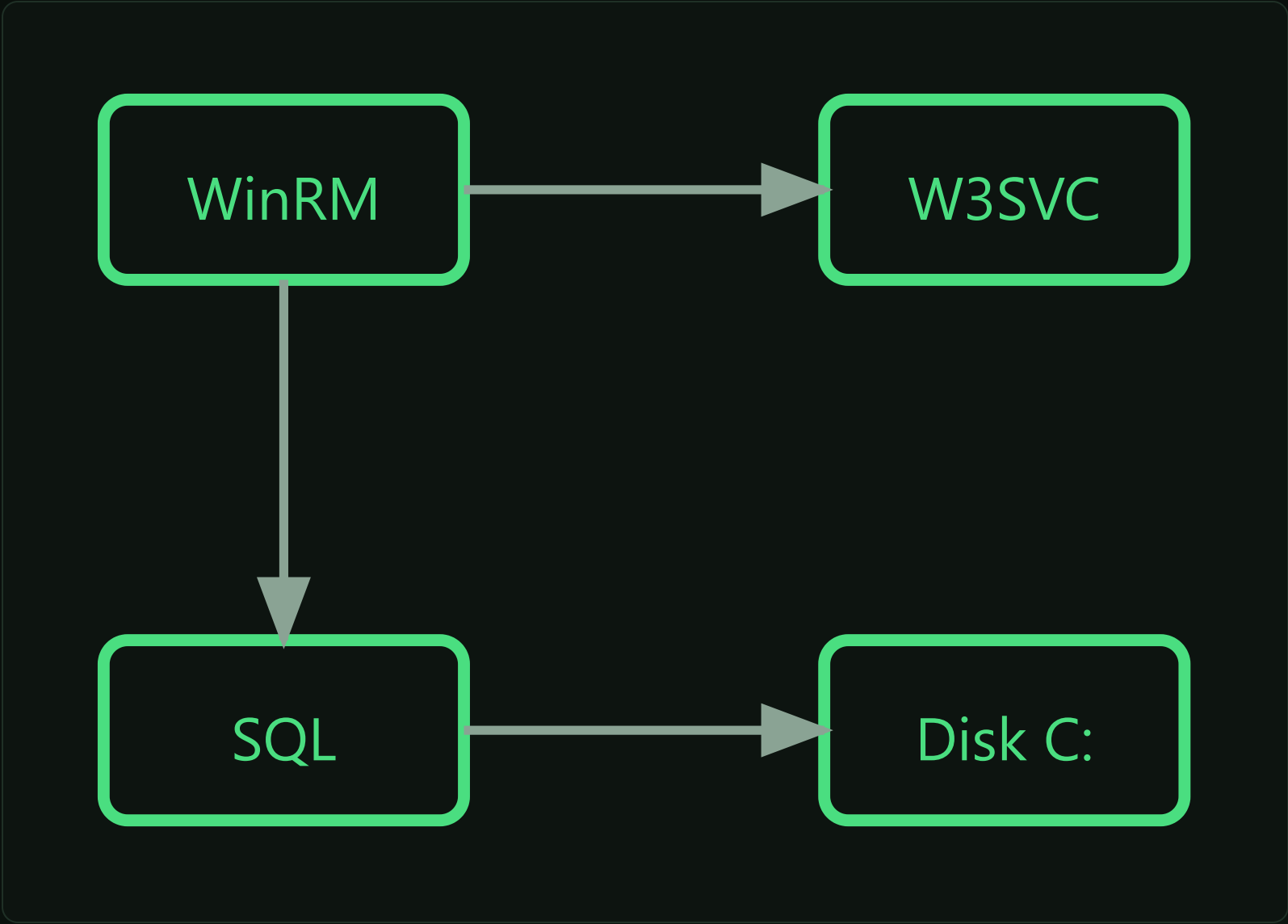
PowerShell MCP + DrawIO MCP = instant architecture docs

POWERSHELL MCP OUTPUT

```
Get-InfraServiceHealth
├ WinRM: Running
├ W3SVC: Running
└ MSSQLSERVER: Running

Get-InfraDiskSpace
└ C: 45% free
```

DRAWIO DIAGRAM



Introspect the system. Draw it. All in English.

Azure Variant (Same Tool, Two Networks)

Same MCP tool. Two **networking models**.

DIRECT (VNET-PEERED)

```
> What's the status of nginx on demo-web-01?
```

```
Transport: Direct  
SSH → 10.42.2.5  
~3-5ms RTT
```

ARCTUNNEL (HYBRID CONNECTIVITY)

```
> And the spooler on ArcBox-Win2K22?
```

```
Transport: ArcTunnel  
Arc SSH proxy  
Network-independent
```

VNet for hosts you provision. Arc for everywhere else. Agent never chooses — host map decides.

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Bridge to M15: Now Let's Talk Guardrails

You've seen agents calling **PowerShell** — discovering scripts, respecting schemas, hitting real servers.

You've seen agents calling **DrawIO MCP** — turning context into diagrams.

Now that agents can introspect AND visualize, let's talk **guardrails**.

M15 (in 10 min): Safety, trusted sources, secrets management, review patterns.

The rules of the Matrix.